TRANSPORTATION MODELS:AN INTRODUCTION

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Some special items are produced at factories at Patna, Guwahati, Mumbai and Hyderabad. Distributors are located at Lucknow, Delhi, Jaipur, Chennai and Kolkata. Supply and demands of items are indicated in table. Note that factories have capacity to produce 18,000 items, but only 15,000 items are in demand. Transportation cost of each item is proportional to distance it travels. The road distances from factory to distribution centers (in km) are indicated in each cell. Your goal is to minimize the transportation cost.

Distributor Factory	Lucknow	Delhi	Jaipur	Chennai	Kolkata	Supply
Patna	610	1,060	1,180	1,490	580	3,000
Guwahati	1,020	1,910	1,946	2,700	1,018	6,000
Mumbai	1,370	1,420	1,150	1,330	2,050	4,000
Hyderabad	1,300	1,560	1,470	630	1,500	5,000
Demand	2,000	7,000	4,000	1,000	1,000	

Unbalanced Problem: add dummy destination

Distributor Factory	Lucknow	Delhi	Jaipur	Chennai	Kolkata	Dummy	Supply
Patna	610	1,060	1,180	1,490	580	0	3,000
Guwahati	1,020	1,910	1,946	2,700	1,018	0	6,000
Mumbai	1,370	1,420	1,150	1,330	2,050	0	4,000
Hyderabad	1,300	1,560	1,470	630	1,500	0	5,000
Demand	2,000	7,000	4,000	1,000	1,000	3000	

Standard problem

X_{11}	X_{12}	…	X_{1n}	a_1
X_{21}	X_{22}	…	X_{2n}	a_2
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
X_{m1}	X_{m2}	…	X_{mn}	a_{2m}
b_1	b_2	.	b_n	

a_i = availablity at supplier i

b_i = demand at destination i

x_{ij} = units supplied from supplier i to destination j

Problem formulation

$$\text{Minimize} \sum_{i=1}^{m}\sum_{j=1}^{n} c_{ij}x_{ij}$$

$$s.t. \sum_{j=1}^{n} x_{ij} = a_i, \quad i = 1,2,...,m$$

$$\sum_{i=1}^{m} x_{ij} = b_j, \quad j = 1,2,...,n$$

$$x_{ij} \geq 0, \quad i = 1,2,...,m;$$

$$j = 1,2,...,n.$$

For consistency,

$$\sum_{i=1}^{m} a_i = \sum_{j=1}^{n} b_j.$$

```
clc;
clear;

C = [610 1060 1180 1300 1420;      % Patna
     750 1910 1980 2100 2020;      % Guwahati
     1290 1230 1150 1330 1550;     % Mumbai
     1350 1420 1470 1600 1500];    % Hyderabad

supply = [3000; 6000; 4000; 5000];            % Supply from each factory
demand = [2000; 7000; 4000; 1000; 1000];      % Demand at each distributor

% ------------------------------
total_supply = sum(supply);
total_demand = sum(demand);

if total_supply > total_demand
    % Add dummy demand point (with zero transportation cost)
    dummy_demand = total_supply - total_demand;
    C = [C, zeros(size(C,1),1)];      % Add dummy column to cost matrix
    demand = [demand; dummy_demand]; % Add dummy demand
elseif total_demand > total_supply
    % Add dummy supply point (with zero transportation cost)
    dummy_supply = total_demand - total_supply;
```

```
 C = [C; zeros(1, size(C,2))];     % Add dummy row to cost matrix
    supply = [supply; dummy_supply]; % Add dummy supply
end

[m, n] = size(C);         % m = #factories, n = #distributors
f = C(:);                 % Flatten cost matrix to column vector (mn x 1)

% ------------------------------
% CONSTRAINTS
% ------------------------------

% Equality constraint matrix Aeq and beq
Aeq = zeros(m+n, m*n);
beq = [supply; demand];

% Supply constraints (rows)
for i = 1:m
    Aeq(i, (i-1)*n + (1:n)) = 1;
end

% Demand constraints (columns)
for j = 1:n
    Aeq(m+j, j:n:end) = 1;
end
```

```
% Solve using linprog
lb = zeros(m*n, 1);        % Lower bounds (no negative transport)
options = optimoptions('linprog', 'Display', 'none');

[x, fval, exitflag] = linprog(f, [], [], Aeq, beq, lb, [], options);

% Output
if exitflag ~= 1
    error('Optimization did not converge.');
end

% Reshape result into m x n matrix
X = reshape(x, [m, n]);

disp('Optimal transportation plan (units from factory to distributor):');
disp(X);
disp(['Minimum total transportation cost: ', num2str(fval)]);
```

Dual of the transportation problem

Primal

$$\text{Minimize} \sum_{i=1}^{m}\sum_{j=1}^{n} c_{ij}x_{ij}$$

$$s.t. \sum_{j=1}^{n} x_{ij} = a_i, \quad i = 1,2,...,m$$

$$\sum_{i=1}^{m} x_{ij} = b_j, \quad j = 1,2,...,n$$

$$x_{ij} \geq 0, \quad i = 1,2,...,m;$$

$$j = 1,2,...,n.$$

For consistency,

$$\sum_{i=1}^{m} a_i = \sum_{j=1}^{n} b_j.$$

Dual

$$\text{Maximize} \sum_{i=1}^{m} a_i u_i + \sum_{j=1}^{n} b_j v_j$$

$$s.t. \quad u_i + v_j \leq c_{ij}, \quad i = 1,2,...,m, \quad j = 1,2,...,n.$$

Primal Constraints

$$x_{11} + x_{12} + ... + x_{1n} = a_1$$

$$x_{21} + x_{22} + ... + x_{2n} = a_2$$

$$.....$$

$$x_{m1} + x_{m2} + ... + x_{mn} = a_m$$

$$x_{11} + x_{21} + x_{m1} = b_1$$

$$x_{12} + x_{22} + ... + x_{m2} = b_2$$

$$.....$$

$$x_{1n} + x_{2n} + x_{nn} = b_n$$

Theorem

A transportation problem always has a solution, but there is exactly one redundant equality constraint. When any one of the redundant constraint is dropped, the remaining system of m+n−1 equality constraint is linearly independent.

Basic feasible solution

Total (m+n−1) x_{ij} are basic. The remaining are non-basic and taken as zero. Some basic variables may be zero. Then we have degenerate basic solution.

We shall discuss three methods of finding basic feasible solutions:

1. Northwest-corner rule
2. Least-cost method
3. Vogel approximation method

Quiz question

- We had to make a dummy destination to balance transportation model. Is there surplus at the source or shortage?
- We had a make dummy source to balance transportation model. Is there surplus at the source or shortage?

Transportation scheme with a dummy destination

	Destination 1	Destination 2	Dummy	Supply
Source 1	100	200	0	3000
Source 2	150	115	0	1000
Source 3	210	70	0	900
Demand	2900	1500	500	

Now, we want to put constraint that source 2 must supply all the items. Hence, put a high number in the second row of dummy column.

1. Formulate the problem in a matrix,
2. Find an initial feasible solution,
3. Optimize the solution to minimize total transportation costs.